

Access Is Not Use: Auditing Causal Bypass of Force in Behavior-Cloned Vision-Language-Action Policies

Anonymous Authors

Paper ID: XXXX

Abstract—Vision-language-action (VLA) policies increasingly receive physical condition signals—wrist wrench, contact, vacuum-seal state—in their observations, yet whether a policy *causally uses* them is unmeasured. On a real-robot suction case-picking task whose critical transition—stop descending on contact—is observable only through force (single head camera; case heights are visually near-indistinguishable), a fine-tuned VLA that receives the full wrench presses through contact until a safety abort in every trial. We introduce *counterfactual authority probes*: a computed condition vector $\hat{c}$ (contact, normal force, seal) injected through zero-initialized FiLM with the raw wrench masked, so that $\hat{c}$ is the only force pathway and the conditioned policy receives *no information the baseline lacks*. Forcing $\hat{c}$ at inference measures how much the condition actually moves the action. The probes expose a *bypass* phenomenon: with the wrench unmasked, conditioning trains to validation loss identical to the fourth decimal yet near-zero authority—a difference invisible to standard validation. A matched quintuple (same data, same initialization, indistinguishable imitation error) shows the architecture does not force usage but *determines its form*: raw access yields a template response that fades as force exceeds the demonstrations, an ungrounded bottleneck yields none, and only the grounded bottleneck brakes monotonically with force—stopping and retreating at 12N, converging in all closed-loop press simulations where the alternatives penetrate. On the robot, force-decorrelated press-retreat demonstrations plus the grounded bottleneck convert 0/3 over-press failures into 5/7 gentle picks (2.5–4.8N), and the dose-response crossover reproduces on-robot. The failure class persists unchanged in a 3.6B-parameter VLA.

Index Terms—imitation learning, vision-language-action models, force control, causal confusion, contact-rich manipulation

I. INTRODUCTION

Behavior-cloned vision-language-action (VLA) policies are rapidly acquiring physical condition signals. Wrist force/torque, contact events, and task-state bits now routinely appear in the observation vector, and a growing line of work fuses force into the policy backbone for contact-rich manipulation [1]–[4]. All of this work shares a validation recipe—add the signal, retrain, report task success—and none of it measures whether the trained policy *causally uses* the signal when it matters: whether, holding everything else fixed, changing the force input changes the action. Imitation learning offers no such guarantee. The objective pins down what the

expert did along the data manifold; *which input the behavior is bound to* is left to whatever correlations the demonstrations happen to contain [5], [6].

The consequence is easiest to see where force is the only signal that can carry the task. We study real-robot suction case picking with a single head-mounted camera and no wrist camera: the arm must stop descending the moment the suction cup meets the case, and stack heights differing by a few millimeters—the entire success margin—are visually near-indistinguishable at head-camera range. The wrench is in the state the policy receives, and every demonstration exhibits the stop. Yet a naively fine-tuned VLA—full wrench in its state, hereafter the *naive policy*—presses through contact in every evaluation trial and never initiates a stop of its own: each run ends only when an external safety layer, separate from the policy, trips at its 15N force limit (15.4–18.7N measured at abort)—the policy still commanding descent. The task turns on a single transition—stop when force rises—and it is precisely this transition that the policy fails to bind to force.

This failure leaves no trace in training. A policy that binds its actions to force and a policy that routes around it can reach the same validation loss—in our experiments, equal to the *fourth decimal place*—so no loss value can certify a fix. We call this routing-around *bypass* and exhibit it directly in Sec. V. A fix must therefore hold *by construction*: leave the policy no route around the signal, and make its use of the signal directly measurable.

Our prescription does both. A condition vector $\hat{c}$ (contact, normal force, seal) is *computed* from signals the naive policy already receives and injected through zero-initialized FiLM modulation [7], and the raw wrench is masked from the state, so $\hat{c}$ becomes the *only* pathway from force to action—no route around. We call this design a *grounded bottleneck*—a bottleneck because $\hat{c}$ is the sole route from force to action, grounded because its content is computed from measured signals—and the policy trained behind it the *conditioned policy*. By construction the conditioned policy receives no information the naive policy lacks; the design changes routing, not content. And because the pathway is explicit, forcing $\hat{c}$ at inference directly measures its *causal authority*—the

fraction of commanded descent the condition can cancel. The demonstrations complete the recipe: each press-retreat episode presses to a randomized force target and retreats before any seal event, so the force at which the expert stops varies from episode to episode—in the data, depth, appearance, and timing no longer predict the stop; only the force reading does. The same interface that fixes the policy is the instrument that audits it (Sec. IV).

The resulting policy is not uniformly better—it is different exactly where the task is decided. In a matched comparison—same data, same initialization, imitation error indistinguishable—the naive policy *does* respond to force at the demonstrated magnitudes, as strongly as the conditioned policy or more so. But its response is a *template*: strongest near what the demonstrations felt, fading as force rises past them. The conditioned policy’s response instead grows with force, to the point of stopping and backing away at forces beyond anything in the training data (Sec. V). The same comparison doubles as an ablation of the recipe: remove the mask and the pathway trains dead (bypass); remove the grounding and the policy becomes the worst in the study. On the robot, the difference appears as *force regulation*: the naive policy’s contact forces are censored at the external abort (15 N+—lower bounds, not chosen stopping forces), the conditioned policy’s sit at 2.5–4.8 N—an order of magnitude below—and live counterfactual probes reproduce the offline crossover on the physical system (Sec. V-E). The failure class also survives scale: a 3.6B-parameter $\pi_{0.5}$ [8] fine-tuned on the same demonstrations shows the same fading-template signature (Sec. V-D).

Our contributions:

- **A recipe for making a VLA use force when it matters.** Force-decorrelated press-retreat demonstrations plus a grounded conditioning bottleneck (computed $\hat{c}$, zero-initialized FiLM, raw-wrench mask) turn the same demonstrations into a force-monotone brake instead of a fading template—on the robot, regulating contact force an order of magnitude below the safety limit on a transition the naive policy fails uniformly (Sec. IV, V-E).
- **An instrument for causal authority.** The bottleneck is measurable by construction: a counterfactual probe battery—condition forcing, state transplants with pathway decomposition, dose-response sweeps, closed-loop press simulation, and on-robot frozen-observation counterfactuals—quantifies moment-resolved causal use and transfers to the real system (Sec. IV-D).
- **Findings standard validation cannot produce.** *Bypass*: with the raw signal available, the conditioning pathway trains causally dead at indistinguishable validation loss—replicated in two independent model pairs—so no quantity in the standard stack detects whether force is used. A matched-quintuple ablation shows each ingredient of the recipe is necessary (mask, grounding, injection site), and dissociates the roles: the data’s correlation structure decides *whether* a signal is used; the conditioning design decides the *form* its use takes under extrapolation. The

failure class persists unchanged at 3.6B scale (Sec. V).

In one sentence: *a policy that stops when it feels what the demonstrations felt and a policy that stops harder the harder it is pressed are indistinguishable in loss and similar at the demonstrated forces—they differ exactly where the task is decided, and only force-decorrelated data behind a grounded, masked FiLM bottleneck yields the second.*

II. RELATED WORK

Force- and tactile-conditioned VLA policies. ForceVLA fuses wrist force as dedicated tokens through a mixture-of-experts backbone [1]; PhaForce schedules policy phases on contact and force signals [2]; ForceFlow [3], FARM [9], and FILIC [4] condition diffusion or impedance-loop policies on force and tactile input; TacForeSight learns force-conditioned tactile prediction [10]. These works *design for* force use and validate it with task success and retraining ablations. Neither measurement establishes causal use at the decision moment: an ablation compares two differently-trained functions, and success can be carried by correlates of force rather than force itself. We do not claim these systems bypass their force inputs—their tasks may contain force information that vision cannot substitute, leaving no shortcut to take. Our claim is narrower and falsifiable: *when* the demonstrations correlate the condition with cheaper cues, a policy with a dedicated force pathway can train to indistinguishable loss while routing around that pathway (we exhibit exactly this with our unmasked control), and no metric in the standard validation stack detects it. We contribute the missing measurement.

Intermediate condition representations. Injecting a compact, task-level representation between observation and action is a recurring design: RT-Affordance conditions on predicted affordance plans [11], and FiLM modulation [7] is the standard mechanism for conditioning a backbone on side information. Our $\hat{c}$ shares the shape but differs in role: it is *computed* from signals the policy already receives—not a learned head that could itself be bypassed—and it is paired with a mask that removes the raw signal, turning the representation from a convenience into a measurable bottleneck.

Causal confusion and shortcut learning. That imitators latch onto spurious correlates is well established: causal confusion in IL [5], copycat behavior on observation histories [12], and shortcut learning in deep networks broadly [6], [13]. Recent work documents shortcuts in generalist robot policies [14] and uses targeted interventions to expose or correct them [15], [16]. We instantiate this literature where it has not been applied—force conditioning in fine-tuned VLAs—and add two elements: a quantitative demonstration that *training loss carries zero signal* about which input the policy is bound to (equal to the fourth decimal in two independent model pairs), and the usage/form dissociation: the data’s correlation structure decides *whether* a signal is used, while the conditioning architecture decides *what form* its use takes under extrapolation.

Privileged supervision and interventional evaluation. Privileged-teacher distillation [17], [18] assumes access to a

signal worth distilling; our probes answer the prior question of whether the deployed policy uses the signal at all. Our press-retreat prescription is a data-side intervention in the spirit of decorrelation fixes for causal confusion [5], but designed from a physical argument (Sec. VI) that reactive gating is arithmetically insufficient on stiff contact, so the demonstrations must teach anticipatory approach and retreat.

III. TASK, SETTING, AND THE FAILURE

Task and platform. We study suction case picking on a dual-arm mobile manipulator: the arm descends onto a stack of rigid cases, must stop when the suction cup meets the top case, hold until vacuum seal is confirmed, and lift (Fig. ??). The margin is tight in both directions—too little press and the seal fails; a few millimeters too much and contact force climbs past the 15N safety limit, since the stack is stiff ($\sim 5\text{N/mm}$, Sec. VI). The policy controls a 7-D end-effector action (Cartesian pose increments and a suction command) at 15Hz, executed in chunks of 5 actions ($\sim 330\text{ms}$ between observations). **[TODO: verify action-space wording (7-D split) against deployment config]**

Observations and their asymmetry. The policy observes a single head-mounted camera (RGB and colorized depth; no wrist camera) and a 15-D state: end-effector position and quaternion, suction command, vacuum-seal bit, and the 6-D wrist wrench. The setting is a deliberately clean diagnostic: at head-camera range, case-stack heights differing by millimeters—the entire success margin—are visually near-indistinguishable (Fig. ??, right), while the wrench and seal bit signal the contact transition directly. Every condition the task depends on is *in the state the policy receives*; the question is whether the policy binds its behavior to it.

Demonstrations. We collect 100 training episodes (19,445 frames) and hold out 6 validation episodes (1,169 frames) with a scripted expert designed to *decorrelate* force from its habitual correlates: each episode descends with suction on, presses to a randomized force target drawn from $[8, 15]\text{N}$, retreats 5–10mm from the contact plane, then hovers and lifts immediately on seal. Every episode thus demonstrates the counterfactual the task requires: force rises $\rightarrow$ stop $\rightarrow$ move back up, *before* any seal event, at a force level that varies independently of depth. This is the data-side prescription whose rationale we develop in Sec. VI; here it matters that the baseline and all conditioned variants are trained on exactly these episodes.

The failure. A SmolVLA policy [19] fine-tuned on these demonstrations—receiving the full wrench and seal bit in its state—fails 3/3 evaluation trials the same way: it descends, makes contact, and keeps pressing until the force limit aborts the run at 15.4–18.7N. The transition that the demonstrations exhibit in every single episode, driven by the one signal that observably announces it, is the transition the policy does not perform. Success rate says the policy is bad; it does not say *why*. The policy might ignore force entirely; it might respond to force but too weakly; it might respond in a form that collapses precisely when force exceeds the demonstrated

range. These hypotheses have opposite prescriptions (more data? different sensors? different architecture?), and none of them is distinguishable from training curves or rollouts. Distinguishing them requires intervening on the signal itself—the instrument we build next.

IV. AN INSTRUMENT FOR CAUSAL AUTHORITY

Our instrument has two halves: a conditioning pathway constructed so that interventions on it are meaningful (IV-A–IV-C), and a battery of counterfactual probes that perform the interventions (IV-D). Throughout, the design goal is *measurement validity*, not architectural novelty: every component exists to make one number well-defined—how much does the condition causally move the action?

A. Computed condition vector $\hat{c}$

From the wrench F and seal bit already present in the state we compute a small set of task-critical conditions—the physical cues a human operator references (“if you feel contact, stop; if the seal is on, lift”):

$$\begin{aligned}\hat{c}_{\text{contact}} &= \text{clip}((\|F\| - F_0) / \tau, 0, 1), \\ \hat{c}_{\text{fmag}} &= (\|F\| - F_0) / \tau_m \quad (\text{unsaturated}), \\ \hat{c}_{f_z} &= (f_z - o_z) / \tau_z, \quad \hat{c}_{\text{seal}} = \text{seal},\end{aligned}\tag{1}$$

with offsets and scales calibrated to the measured force statistics of the demonstrations (e.g., F_0 at the contact-onset magnitude).¹ Two properties are essential. *Computed, not learned:* $\hat{c}$ is a fixed function of observed signals, so it cannot itself be bypassed during training the way a learned condition head could. *Zero new information:* every channel is a deterministic function of state dimensions the baseline also receives, so conditioning changes *routing*, never *content*—the “it works because it got extra sensing” objection is excluded by construction. The channels are deliberately plural: keeping contact, force magnitude, and seal separate is what later lets the probes attribute authority to specific signals (Sec. V-B).

B. Zero-initialized FiLM injection

$\hat{c}$ enters the policy through FiLM modulation [7], with the modulation initialized to identity so that training starts from the unconditioned policy. We instantiate two injection points in SmolVLA: *prefix* (modulating the state token entering the VLM backbone) and *suffix* (modulating the action-expert input). The comparison is not incidental—the two sites differ consistently in the authority they acquire (Sec. V-A).

C. The force-mask bottleneck

With $\hat{c}$ injected but the raw wrench still present in the state, the policy has two routes to force. Masking the wrench dimensions from the state closes the raw route, making $\hat{c}$ the *only* pathway from force to action. The mask is the

¹Calibration quality matters and is itself a finding: the four-model matrix of Sec. V-A uses an earlier, coarser calibration; the matched quintuple of Sec. V-C uses offsets re-estimated from the measured contact distribution plus the unsaturated $\hat{c}_{\text{fmag}}$ channel. An uncalibrated transplant onto a different architecture fails outright (Sec. VI).

instrument’s validity device: under it, forcing $\hat{c}$ intervenes on the totality of the policy’s force access, so measured authority is a property of the policy’s force use, not of one redundant pathway among several. It also strictly *reduces* the conditioned model’s information relative to the baseline. The unmasked configuration is retained as a control—and turns out to demonstrate bypass (Sec. V-A).

D. Counterfactual probe battery

All probes intervene on a trained, frozen policy; none involves retraining. We define **authority** as the intervention-induced change in the commanded vertical action on committed-descent frames, expressed as a percentage of the mean commanded descent: 0% means the intervention is ignored, 100% means it fully cancels descent, >100% means reversal (retreat).

(P1) Condition forcing (open loop). With images and state frozen, $\hat{c}$ is forced from its observed value to a target pattern—either all channels saturated (*full forcing*) or a calibrated contact-onset pattern matching measured first-contact statistics (*contact-moment forcing*). Applicable to conditioned models only.

(P2) State transplant (open loop). The force-bearing dimensions of a *measured* first-contact state are transplanted into pre-contact descent frames of held-out episodes. This probe applies to any policy, including the baseline, and supports a pathway decomposition: apply the transplant to the raw state only (d_{raw}), to the $\hat{c}$ computation only (d_{FiLM}), or to both (total), attributing the response to the raw route versus the conditioning route. [TODO: verify exact transplanted dims (wrench+seal vs full state) in probe_state_authority.py before submission]

(P3) Dose-response sweep. The transplanted wrench is rescaled to target magnitudes (8, 10, 12 N), tracing the policy’s response as a function of force *dose*. The demonstrations peak near the low end of this range, so the sweep measures extrapolation: a policy that has bound braking to force should brake *more* at higher dose; one that has memorized a contact template should respond most near the template and fade beyond it. The sweep’s *slope sign* is the paper’s central discriminator.

(P4) Closed-loop press simulation. Open-loop probes cannot show consequences. We place the frozen policy in a closed loop on held-out episodes: commanded descent advances a virtual penetration depth, a linear-stiffness force model raises the simulated wrench accordingly, and the seal event is withheld—simulating the operationally critical misalignment scenario in which no press depth ever produces a seal. Outcomes: whether the policy stops before the force limit, and maximum penetration. A control variant grants the seal at a shallow depth, separating force-driven stopping from seal-event-driven stopping.

(P5) Live counterfactual probe. The full loop contains physics no offline probe reproduces. On the robot, we freeze observations at staged poses along the approach and apply (P1)–(P3) interventions to *two* policies—the conditioned

TABLE I
BYPASS MATRIX: VALIDATION LOSS VS. CAUSAL AUTHORITY (VAL-BEST / LAST CHECKPOINTS). FULL FORCING = P1 ALL-CHANNEL;
CONTACT-MOMENT = P1 CALIBRATED ONSET PATTERN.

Policy	Val loss (best)	Authority (best / last)	
		full forcing	contact-moment
baseline (raw wrench)	0.15087	—	—
FiLM prefix, wrench kept	0.15075	7/8%	3/4%
FiLM prefix, wrench masked	0.14624	76/54%	7/8%
FiLM suffix, wrench masked	0.15924	60/42%	10/1%

model and the baseline—on identical frozen observations, so both receive the same physical counterfactual. Two operational details matter for validity: force/torque baseline drift is re-anchored at a pre-descent hover pose before probing (drift of up to ~ 1 N within a day was measured, pose-sensitive), and absolute transplant targets are shifted by the measured drift so that both models receive the same effective dose.

Scope. P1–P3 are single-frame and open-loop; P4 closes the loop under a force model; P5 closes the gap to the real system. No single probe is decisive—the findings below rest on their agreement.

V. EXPERIMENTS

We present the offline probe battery first (V-A–V-D), then the robot validation (Sec. V-E–V-F). All offline probes run on the six held-out validation episodes. Checkpoints are selected by validation loss (*best*); the final checkpoint (*last*) is reported where it qualifies the story.

A. Bypass: standard validation cannot see causal use

Table I crosses injection site (prefix/suffix) and wrench masking (on/off) for policies trained identically on the same data.

Three observations. **(i) Bypass.** With the wrench left in the state, the FiLM-conditioned policy trains to a validation loss equal to the baseline’s in the *fourth decimal place* (0.15075 vs. 0.15087) while its condition channels carry 7–8% authority: gradient descent satisfied the imitation objective through the raw route and left the dedicated conditioning pathway causally dead. Nothing in the training signal distinguishes this policy from one that uses its conditions. **(ii) Loss rank anti-correlates with authority rank.** The masked prefix model has both the *lowest* loss and the highest authority; the suffix model has the *worst* loss yet 60% authority; the unmasked model sits between them in loss with none. A practitioner selecting on loss learns nothing about causal use (Fig. ??). **(iii) Selection and site effects.** Validation-loss checkpoint selection also selects for authority (best > last in every conditioned model), and prefix injection dominates suffix. The bypass result replicates in an independent pair below (Sec. V-C): warm-started variants with and without the mask again differ by 10^{-4} in loss and categorically in authority.

B. Where authority binds: stable states, not the transient

Authority, where it exists, is not uniformly distributed over the condition channels. Contact-moment forcing (P1, calibrated onset pattern) yields only 7–10% even for the best-conditioned model (Table I, last column): the 1–2-frame contact *transient* is weakly adopted across the board. The state-transplant probe (P2) shows what is adopted instead. Transplanting a measured first-contact state into pre-contact descent frames moves the *baseline* by 105% of its descent command—the baseline is not force-blind; its raw route responds strongly to force-bearing state. But transplanting a *sealed* state moves both models far more than the contact state does (on-robot: 147% vs. 67% for the baseline; Sec. V-E), and the strongest single responses everywhere are to the stable, sustained signals—seal and sustained press—rather than to contact onset. [TODO: add contact-z distribution analysis from 0729 dataset (bimodal contact heights) — derivation pending, Fig. ?? placeholder]

The picture: what a policy adopts is decided by the *statistical character* of the signal. Stable, persistent conditions (seal state) are adopted essentially for free; a transient that is redundant with depth-like cues in ordinary demonstrations is not. This has an operational corollary developed in Sec. VI: on stiff contact, even a policy with substantial condition authority cannot succeed by *reactive* gating at the contact moment—the successful mechanism is anticipatory behavior plus post-hoc state gating, which is exactly what the probes measure as adopted.

C. The matched quintuple: architecture determines the form of force use

The section’s central experiment holds everything fixed except the conditioning pathway. Starting from the *same* trained baseline checkpoint, on the *same* demonstrations, we continue training four variants: the baseline itself, FiLM *without* the mask (bypass control), FiLM with mask but *shuffled* $\hat{c}$ during training (ungrounded control: the bottleneck exists, its content is noise), and FiLM with mask and grounded $\hat{c}$. Imitation error on held-out episodes is indistinguishable across all five policies (0.81–0.96 mm per step): on the data manifold, these are the same policy. Under intervention they are three different mechanisms (Table II, Fig. ??).

Grounding, not capacity, is the active ingredient. The ungrounded control kills every alternative explanation at once: it has the same added parameters, the same continued training, the same bottleneck—and zero response at every dose, with the worst closed-loop outcome in the study (431 mm penetration). An ungrounded bottleneck is *worse* than raw access: the mask removed the raw route and the noise-trained $\hat{c}$ replaced it with nothing. Conversely, the unmasked control shows that adding the FiLM module without the mask changes nothing: its dose-response curve lies on the baseline’s (+1.57 $\rightarrow$ +1.12 vs. +1.41 $\rightarrow$ +1.12; identical at 12 N), its pathway decomposition puts the response entirely on the raw route, and its loss matches the grounded variant’s to the fourth decimal (0.14750

vs. 0.14762)—the bypass finding, replicated with authority now at 94% vs. 0.

The form dissociation. The baseline responds to force—at the demonstrated magnitude. As the dose rises past the demonstrations, its response *fades* (36% $\rightarrow$ 28%): a template, matched ever more poorly as reality departs from it. The grounded bottleneck’s response *grows monotonically*, crossing full cancellation into retreat at 12 N—where the unsaturated channel reads five times its training range. The monotone extrapolation is not learned from data (no demonstration presses to these forces); it is inherited from the channel’s design as a calibrated monotone scalar. Same data, same imitation: raw access binds a template, the grounded bottleneck binds a brake.

Honest boundaries of the discriminator. The baseline’s *shape* is checkpoint-dependent: at 50k steps its curve rises mildly (57% $\rightarrow$ 65%). The robust discriminators are therefore not slope sign alone but (i) slope *magnitude* ($\sim 10\times$: +2.4–2.5 mm vs. ± 0.3 mm over the sweep), (ii) whether the curve *crosses* full cancellation (only the grounded models stop and retreat; no baseline checkpoint exceeds 65%), and (iii) the closed-loop consequence—the late baseline’s larger open-loop response coexists with a *worse* press-sim outcome (3/6, max 16.0 mm), because a response that saturates below 100% cannot stop a stiff press regardless of its size. The grounded form is stable across checkpoints (6/6 at both). Finally, the seal-granted control variant of P4 localizes the baseline’s failure: given a seal event at shallow depth it stops normally (0.5 mm over-travel); withhold the seal and it presses through. Its stopping is real but bound to the *seal event*; the grounded policy’s is bound to *force*.

D. Scale does not change the class

Fine-tuning $\pi_{0.5}$ (3.6B parameters, $8\times$ SmolVLA) [8] on the same demonstrations reproduces the baseline signature: strong on-manifold transplant response (65–97%), a dose-response that saturates shallowly (never exceeding 37% cancellation, approaching no full stop at any dose), and closed-loop collapse in the seal-never press simulation (0/6, max penetration 86 mm—worse than the smaller model), at equal-or-better imitation error (0.69–0.89 mm). Neither parameter count nor a different VLA architecture converts template responding into force-bound braking; the failure class is a property of the objective-plus-data, not of model scale.

E. Robot validation: closed-loop picks

Robot results are a small- n case study in a controlled diagnostic setting; their role is to test whether the mechanisms measured offline—template fading, monotone braking, the crossover—survive contact with the real closed loop, not to establish success statistics. All runs use a 15 N force limit and 5-step action chunks.

²One prefix failure was a light press that never sealed; the single over-press run used a different checkpoint revision than the other six—per-revision accounting is 5/6 for the primary checkpoint. [TODO: finalize revision pinning before submission]

TABLE II

MATCHED QUINTUPLE: SAME DATA, SAME INITIALIZATION, INDISTINGUISHABLE IMITATION—THREE MECHANISMS. DOSE-RESPONSE = P3 CANCELLATION AT 8 N→12 N (% OF OWN DESCENT). PRESS SIM = P4 SEAL-NEVER: EPISODES STOPPED BEFORE FORCE LIMIT / MAX PENETRATION.

Policy	Force access	Dose-response 8→12 N	Transplant (P2)	Press sim (P4)	Val err
baseline (best)	raw wrench	+1.41 → +1.12 mm (36→28%) ↓	+1.63 (105%)	4–5/6, max 14.9 mm	0.84 mm
baseline (last)	raw wrench	+1.73 → +2.00 mm (57→65%)	+1.69 (108%)	3/6, max 16.0 mm	0.71 mm
FiLM, wrench kept	raw + dead $\hat{c}$	+1.57 → +1.12 mm (d_{raw} carries all)	+1.68 ($d_{\text{FiLM}}=0.04$)	4–5/6, max 11.9 mm	0.85 mm
FiLM, mask, shuffled $\hat{c}$	ungrounded bottleneck	−0.06 → −0.11 mm (≈ 0)	−0.06 (0%)	0/6, max 431 mm	0.96 mm
FiLM, mask, grounded (best)	grounded bottleneck	+1.20 → +4.40 mm (31→115%) ↑	+1.57 (94%, all d_{FiLM})	6/6, max 4.5 mm	0.87 mm
FiLM, mask, grounded (last)	grounded bottleneck	+0.97 → +3.44 mm ↑	+1.45 (97%)	6/6, max 4.1 mm	0.81 mm

TABLE III

ROBOT CASE-PICK OUTCOMES (SAME DEMONSTRATIONS FOR ALL POLICIES).

Policy	Success	Over-press	Contact force
baseline	0/3	3/3 (15.4–18.7 N)	—
FiLM prefix, mask	5/7	1/7 ²	2.5–4.8 N
FiLM suffix, mask	1/3	2/3 (16.6/19.5 N)	6.9 N

The baseline fails uniformly by its offline signature: contact, continued descent, force-limit abort—the closed-loop realization of a response that saturates below full cancellation. The grounded prefix policy converts this to 5/7 success with contact forces of 2.5–4.8 N, an order of magnitude below the abort threshold, sealing in 14–41 s and lifting cleanly. The suffix policy lands between, consistent with its intermediate offline authority. During successful runs the FiLM modulation is measurably active at the transitions ($|\gamma|$ +48%, $|\beta|$ +79% at contact/seal relative to descent).

F. Robot validation: on-robot counterfactuals

The live probe (P5) freezes observations at staged poses and doses both policies with identical counterfactual states. Two sessions matter (Fig. ??).

Fair dose. At transplant doses matching the demonstrations ($\leq$ contact magnitude), the *baseline* responds as strongly or stronger than the conditioned policy (first-contact: 67% vs. 32% of own descent; sealed state: 147% vs. 58%)—in quantitative agreement with the offline dose-response curves at the same doses, where the template peaks. The conditioned policy’s response is dose-monotone across the staged conditions (hover +0.29 → pre-seal +0.87 → sealed +1.41 → +6 N +1.70 mm/frame).

High dose. Raising the forced normal force to +6/ +9/ +12 N reproduces the offline crossover on the robot: the conditioned policy accelerates its braking, +1.19 → +4.59 → +8.56 mm/frame—at 12 N, twice its own descent command, i.e., commanded retreat—while the baseline’s response collapses at the margin (+1.34 → +3.18 → +4.18; increments +1.84 → +1.00). The reversal sits between 6 and 9 N. At near-surface poses the separation is starkest: baseline +1.0–1.8 vs. conditioned +7.5 mm/frame at 12 N.

Isn’t the baseline better, then? At mid dose, yes—and it does not matter. Over-press is a dynamical process in which force rises *past* the demonstrated magnitude; what determines

the outcome is the response exactly where the template fades. The baseline’s brake weakens as it becomes more necessary (closed loop: 0/3; press-sim penetration to the force limit), while the conditioned brake strengthens (5/7; 6/6 stops in simulation). The baseline’s strongest counterfactual response is to the *sealed* state—but over-press is precisely the scenario in which the seal never arrives: its most powerful brake is bound to an absent signal. A policy that stops “when it feels exactly what the demonstrations felt” and a policy that stops “harder the harder it is pressed” are indistinguishable in loss, similar at fair dose, and categorically different where the task is decided.

Two validity notes: force-torque baselines were re-anchored at a pre-descent hover before each session, and absolute transplant targets were drift-shifted so both policies received the same physical dose; live conditioned-policy magnitudes are conservative lower bounds, as some frozen states are slightly off-manifold for its input distribution.

VI. DISCUSSION AND LIMITATIONS

Why the prescription must be anticipatory. A first-principles calculation explains both the failure and the fix. The case stack is stiff: ~ 5 N/mm of penetration. Stopping below the 15 N limit therefore requires a full stop within ~ 2 mm of contact—under 330 ms action chunks and the demonstrated descent rates, this is arithmetically impossible for *reactive* gating that begins braking at the contact transient, no matter how much authority the transient carries. The viable mechanisms are anticipatory: approach slowly, press deliberately, retreat. The press-retreat demonstrations teach exactly this behavior prior, and the probes confirm the resulting division of labor: the successful policy’s contact-moment authority is small (7%) while its full-condition authority is large (76%)—success comes from an anticipatory behavior prior plus post-hoc state gating, not from a reflex. The instrument’s value here is not that it found a large number, but that it correctly *predicted which mechanism could not work*.

Grounding is not free. Transplanting the instrument naively fails, and the failure is diagnostic. Attaching the same conditioning to $\pi_{0.5}$ via suffix injection with uncalibrated off-sets produces the ungrounded-control signature at scale: zero authority at every dose—including with the contact channel forced to saturation—and closed-loop penetration *worse* than the raw $\pi_{0.5}$ baseline (0/6, max 94–102 mm vs. 86 mm), at unchanged imitation error. The mask removes the raw route;

if $\hat{c}$ fails to ground, what remains is a policy with genuinely no force pathway. A bottleneck plus counterfactual data is necessary but not sufficient: injection site and calibration must be re-engineered per architecture (consistent with prefix > suffix and the calibration sensitivity observed within SmolVLA). [DECISION: keep this paragraph’s numeric detail vs. compress to two sentences — page budget]

Operational asymmetry. The computed condition creates knobs the baseline lacks. Force-torque baselines drift (~ 1 N within a day, pose-sensitive); the conditioned policy re-anchors its offsets at a hover pose each run—a deployment-time correction with no retraining—while the baseline’s force statistics are frozen in its normalization. The same interface is what makes the policy auditable (P1/P5 probes) and intervenable at deployment.

Limitations. Single task, single platform, small robot- n : the robot results are a case study, and the statistical claims live in the offline battery ($n=6$ held-out episodes $\times$ dense frames, five matched policies, two architectures). The press simulation depends on a linear force model; we mitigate with the seal-granted control and the live probes, which bracket the simulated mechanism from both sides. Contact-moment (reactive) authority plateaus low in every configuration we trained—in development phases, oversampling transition frames also failed to move it; distilling an explicit descend-until-contact oracle (e.g., via DAGger [17], [20]) is the natural next step, as is a negative-control task whose critical condition is *not* observable in the state, and replication of the bypass audit on force-fusion architectures whose designs assume use [1], [3].

VII. CONCLUSION

On a task whose critical transition is observable only through force, we showed that a behavior-cloned VLA policy given the full signal does not bind its behavior to it—and that no number in the standard validation stack could have revealed this. A computed condition vector behind a zero-new-information FiLM bottleneck turns the question into a measurement: counterfactual authority probes expose bypass at identical loss, localize authority to stable post-hoc states rather than the contact transient, and—in a matched quintuple with indistinguishable imitation error—show that the data’s correlation structure decides *whether* force is used while the conditioning design decides the *form* its use takes: a template that fades beyond the demonstrations, or a calibrated brake that strengthens monotonically, stops, and retreats. The distinction is invisible on the manifold and decisive off it, on the robot as offline. Imitation pins down the trajectory. If we want the mechanism pinned too, we must give the policy demonstrations that decorrelate the condition from its correlates—and an interface through which the condition’s use can be shaped, measured, and corrected.

REFERENCES

[1] TODO-verify authors, “ForceVLA: Force-aware mixture-of-experts for contact-rich manipulation,” *arXiv preprint arXiv:2505.22159*, 2025.

[2] Wang et al. (TODO-verify), “PhaForce: Phase-scheduled visual-force policy learning with slow planning and fast correction,” *arXiv preprint arXiv:2603.08342*, 2026.

[3] TODO-verify authors, “ForceFlow (todo-verify full title),” *arXiv preprint arXiv:2605.11048*, 2026.

[4] —, “FILIC: Dual-loop force-guided imitation learning with impedance control (todo-verify),” *arXiv preprint arXiv:2509.17053*, 2025.

[5] P. de Haan, D. Jayaraman, and S. Levine, “Causal confusion in imitation learning,” in *NeurIPS*, 2019, arXiv:1905.11979.

[6] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.

[7] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, “FiLM: Visual reasoning with a general conditioning layer,” in *AAAI*, 2018.

[8] K. Black *et al.*, “ $\pi_{0.5}$: A vision-language-action model with open-world generalization,” *arXiv preprint arXiv:2504.16054*, 2025, tODO-verify arXiv ID and author list.

[9] TODO-verify authors, “FARM: Tactile-conditioned diffusion policies (todo-verify full title),” *arXiv preprint arXiv:2510.13324*, 2025.

[10] —, “TacForeSight: Force-conditioned tactile world models (todo-verify full title),” *arXiv preprint arXiv:2606.11184*, 2026.

[11] S. Nasiriany *et al.*, “RT-Affordance: Affordances are versatile intermediate representations for robot manipulation,” *arXiv preprint arXiv:2411.02704*, 2024.

[12] C. Wen, J. Lin, T. Darrell, D. Jayaraman, and Y. Gao, “Fighting copycat agents in behavioral cloning from observation histories,” in *NeurIPS*, 2020, arXiv:2010.14876.

[13] K. Hermann, H. Mobahi, T. Fel, and M. C. Mozer, “On the foundations of shortcut learning,” *arXiv preprint arXiv:2310.16228*, 2023.

[14] TODO-verify authors, “Shortcut learning in generalist robot policies (todo-verify full title),” *arXiv preprint arXiv:2508.06426*, 2025.

[15] —, “Intervention-based evaluation of imitation policies (todo-verify full title),” *arXiv preprint arXiv:2507.22380*, 2025.

[16] —, “Interventional imitation learning (todo-verify full title),” *arXiv preprint arXiv:2307.15980*, 2023.

[17] D. Chen, B. Zhou, V. Koltun, and P. Krähenbühl, “Learning by cheating,” in *CoRL*, 2019.

[18] TODO-verify authors, “Learning long-horizon robot manipulation skills via privileged action,” *arXiv preprint arXiv:2502.15442*, 2025.

[19] M. Shukor *et al.*, “SmolVLA: A vision-language-action model for affordable and efficient robotics,” *arXiv preprint arXiv:2506.01844*, 2025, tODO-verify arXiv ID.

[20] S. Ross, G. J. Gordon, and J. A. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *AISTATS*, 2011.